

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                    |
|----------------------|--------------------|
| United States Patent | 12418734           |
| Kind Code            | B2                 |
| Date of Patent       | September 16, 2025 |
| Inventor(s)          | Ma; Jiaju et al.   |

---

### Global shutter pixel with vertically integrated multi-phase charge transfer

---

#### Abstract

An image sensor may include a plurality of pixels, each of which may include a photodiode having a charge accumulation region (“PD”), a floating diffusion region (“FD”), and a charge transfer region vertically between the PD and FD. The vertical charge transfer region may include a first charge modulation region (“P1”), a second charge modulation region (“P2”), and a third charge modulation region (“P3”). The image sensor may operate in a global shutter mode, in which the P2 may be used as an in-pixel charge memory region to temporarily store charge during transfer of the charge from PD to FD via P1, P2, and P3.

---

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| <b>Inventors:</b> | <b>Ma; Jiaju (Arcadia, CA), Laflaquiere; Arnaud (Paris, FR)</b> |
| <b>Applicant:</b> | <b>Apple Inc. (Cupertino, CA)</b>                               |
| <b>Family ID:</b> | <b>1000008820102</b>                                            |
| <b>Assignee:</b>  | <b>Apple Inc. (Cupertino, CA)</b>                               |
| <b>Appl. No.:</b> | <b>18/326940</b>                                                |
| <b>Filed:</b>     | <b>May 31, 2023</b>                                             |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240236525 A1          | Jul. 11, 2024           |

#### Related U.S. Application Data

us-provisional-application US 63479533 20230111

---

#### Publication Classification

Int. Cl.: H04N25/75 (20230101)

U.S. Cl.:

CPC H04N25/75 (20230101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl.   | CPC              |
|--------------|-------------|---------------|------------|------------------|
| 6046466      | 12/1999     | Ishida        | 348/E5.037 | H01L<br>27/14609 |
| 8274587      | 12/2011     | Hynecek       | N/A        | N/A              |
| 2020/0068155 | 12/2019     | Kitano        | N/A        | H04N 25/77       |
| 2021/0296378 | 12/2020     | Fukui et al.  | N/A        | N/A              |
| 2021/0408096 | 12/2020     | Shimizu       | N/A        | N/A              |
| 2022/0320174 | 12/2021     | Yang et al.   | N/A        | N/A              |

FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 104241311  | 12/2013          | CN      | N/A |
| 2009066909 | 12/2008          | WO      | N/A |

OTHER PUBLICATIONS

International Search Report and Written Opinion from PCT.US2023/086492, dated Mar. 25, 2024, pp. 1-14. cited by applicant

Primary Examiner: Monk; Mark T

Attorney, Agent or Firm: Kowert, Hood, Munyon, Rankin & Goetzel, P.C.

Background/Summary

(1) This application claims benefit of priority of U.S. Provisional Application Ser. No. 63/479,533, entitled “Global Shutter Pixel with Vertically Integrated Multi-Phase Charge Transfer”, filed Jan. 11, 2023, which is hereby incorporated in reference herein in its entirety.

BACKGROUND

Technical Field

(1) This disclosure relates generally to an image sensor and more specifically to pixels of an image sensor having vertically integrated multi-phase charge transfer for image capturing in a global shutter mode.

Description of the Related Art

(2) Image capturing devices, such as cameras, are widely used in various electronic devices, such

as mobile devices (e.g., smart phones, tablets, laptops, etc.), robotic equipment, or security monitoring devices, among others. An image capturing device may include an image sensor having a plurality of light-gathering pixels. Each pixel may include a photodiode. The image capturing device may capture light from an environment and pass the light to the image sensor. When exposed to light, photodiodes of the pixels may accumulate electrical charge. At readout, the electrical charge of the photodiodes may be read out of the photodiodes, using one or more transistors, to generate analog image signals. The analog image signals may be converted to digital signals and further processed to produce images.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows top view and cross-section view of an example pixel of an image sensor, according to some embodiments.
- (2) FIG. 2 shows example implementations for control gates of a pixel of an image sensor, according to some embodiments.
- (3) FIG. 3 shows example implementations for vertical gate areas of a pixel of an image sensor, according to some embodiments.
- (4) FIGS. 4A-4D show example timing diagrams of control signals and potential profile of a pixel of an image sensor to illustrate charge transfer, according to some embodiments.
- (5) FIG. 5A shows a cross-section view of an example pixel and a top view of an image sensor, according to some embodiments.
- (6) FIG. 5B shows example control signals applied to different components of a plurality of pixels of an image sensor, according to some embodiments.
- (7) FIG. 6 shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments.
- (8) FIG. 7 shows example isolation structures of a pixel of an image sensor, according to some embodiments.
- (9) FIG. 8 shows an example split pixel for autofocus application, according to some embodiments.
- (10) FIG. 9 shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments.
- (11) FIG. 10 shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments.
- (12) FIG. 11 shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments.
- (13) FIG. 12 is a block diagram of an example image capturing device, according to some embodiments.
- (14) FIG. 13 is a flowchart showing an example method for using an in-pixel charge memory region to operate an image sensor in a global shutter mode, according to some embodiments.
- (15) FIG. 14 illustrates a schematic representation of an example device that may include an image capturing device (e.g., a camera) having an image sensor that includes pixels with disclosed in-pixel charge memory regions, according to some embodiments.
- (16) FIG. 15 illustrates a schematic block diagram of an example computing device, referred to as computer system, that may include or host embodiments of an image capturing device (e.g., a camera) having an image sensor that includes pixels with disclosed in-pixel charge memory regions, according to some embodiments.
- (17) This specification includes references to “one embodiment” or “an embodiment.” The appearances of the phrases “in one embodiment” or “in an embodiment” do not necessarily refer to the same embodiment. Particular features, structures, or characteristics may be combined in any

suitable manner consistent with this disclosure.

(18) “Comprising.” This term is open-ended. As used in the appended claims, this term does not foreclose additional structure or steps. Consider a claim that recites: “An apparatus comprising one or more processor units.” Such a claim does not foreclose the apparatus from including additional components (e.g., a network interface unit, graphics circuitry, etc.).

(19) “Configured To.” Various units, circuits, or other components may be described or claimed as “configured to” perform a task or tasks. In such contexts, “configured to” is used to connote structure by indicating that the units/circuits/components include structure (e.g., circuitry) that performs those task or tasks during operation. As such, the unit/circuit/component can be said to be configured to perform the task even when the specified unit/circuit/component is not currently operational (e.g., is not on). The units/circuits/components used with the “configured to” language include hardware—for example, circuits, memory storing program instructions executable to implement the operation, etc. Reciting that a unit/circuit/component is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) for that unit/circuit/component. Additionally, “configured to” can include generic structure (e.g., generic circuitry) that is manipulated by software and/or firmware (e.g., an FPGA or a general-purpose processor executing software) to operate in manner that is capable of performing the task(s) at issue. “Configure to” may also include adapting a manufacturing process (e.g., a semiconductor fabrication facility) to fabricate devices (e.g., integrated circuits) that are adapted to implement or perform one or more tasks.

(20) “First,” “Second,” etc. As used herein, these terms are used as labels for nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.). For example, a buffer circuit may be described herein as performing write operations for “first” and “second” values. The terms “first” and “second” do not necessarily imply that the first value must be written before the second value.

(21) “Based On.” As used herein, this term is used to describe one or more factors that affect a determination. This term does not foreclose additional factors that may affect a determination. That is, a determination may be solely based on those factors or based, at least in part, on those factors. Consider the phrase “determine A based on B.” While in this case, B is a factor that affects the determination of A, such a phrase does not foreclose the determination of A from also being based on C. In other instances, A may be determined based solely on B.

(22) It will also be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the intended scope. The first contact and the second contact are both contacts, but they are not the same contact.

(23) The terminology used in the description herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(24) As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon

determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

#### DETAILED DESCRIPTION

(25) Various embodiments described herein relate to an image sensor operating in a global shutter mode. In some embodiments, the image sensor may include a plurality of light-gathering pixels, e.g., organized in a pixel array having one or more rows and one or more columns of pixels. In some embodiments, the image sensor may be a CMOS (Complementary Metal Oxide Semiconductor) image sensor, a CCD (charge-coupled device) image sensor, and the like. In some embodiments, the image sensor may be part of an image capturing device, e.g., a camera, which further may be part of an electronic device, e.g., a mobile device (e.g., smart phones, tablets, laptops, etc.), robotic equipment, or a security monitoring device, among others. In some embodiments, the pixels of the image sensor may each include at least one photodiode comprising a charge accumulation region (hereinafter “PD”), a floating diffusion region (hereinafter “FD”), and a charge transfer region vertically between the PD and FD. When exposed to light, for each pixel, the PD may accumulate charge or photo-carriers. At readout, at least some of the charge may be transferred from the PD to the FD to generate an analog image signal (e.g., an analog voltage) at the FD, which may be further accessed at a pixel output line outside the pixel. In some embodiments, the analog image signals accessed through pixel output lines may be further processed, e.g., analog-to-digital converted using analog-to-digital converter(s) and digitally processed by an image signal processor (ISP), to generate one or more images.

(26) Generally, a given image capturing device may operate in a rolling shutter mode or a global shutter mode. In rolling shutter, different lines of the pixel array of the image sensor of the image capturing device may be exposed to light at different times as the read out “wave” sweeps through the image sensor. For example, the pixels of a pixel array may be exposed and image signals of the pixels may be read out sequentially, e.g., row-by-row from the top to the bottom of the pixel array. For example, pixels of the same row may be read out at the same time, whereas pixels in the same column but different rows may be read out sequentially one by one. Thus, in rolling shutter, the image sensor may record an image row-by-row sequentially instead of capturing the entire image at once. By comparison, in global shutter, all the pixels may have the same exposure time, meaning that each pixel in the image sensor may begin and end exposure simultaneously. As a result, the entire image may be recorded at once. The rolling shutter can cause color and/or shade variation in the captured images, since different “lines” of the image are recorded at different times. In some application, e.g., in high-speed photographing or recording, this can cause severe distortions and greatly affect qualities of the captured images. Thus, in some embodiments, global shutter may be preferred. However, in some embodiments, even though the pixels of an image sensor end exposure simultaneously, their image signals may still be read out sequentially, e.g., row-by-row, like the rolling shutter. Thus, the image sensor may need “memory” to temporarily store (a) the charge (e.g., in the charge domain) and/or (b) the analog or digital image signals (e.g., in the voltage domain) of the pixels at the end of exposure until readout of the individual pixels.

(27) To solve the problem, in some embodiments, the pixels of the image sensor disclosed herein may each include an in-pixel charge memory region. At the end of exposure, the charge may be transferred from the PD to the in-pixel charge memory region. The charge may be temporarily stored there until readout of the pixel. At the readout, the charge may be transferred from the in-pixel charge memory region to the FD, from which analog image information may be further accessed through a pixel output line. One with skills in the art shall understand that the disclosed image sensor can provide several benefits. First, it can provide an in-pixel charge memory region for each pixel to temporarily store the pixel's charge, thus enabling an image sensor to operate in a global shutter mode. In addition, the “memory” is integrated as part of the pixel to store the charge inside the pixel, thus eliminating or at least reducing other additional memory storage components (e.g., memory chips on the image sensor, etc.). As a result, this may reduce the number of

components for an image sensor, reduce the image sensor's footprint, and/or increase the sensor's pixel density.

(28) FIG. 1 shows top and cross-section view of an example pixel of an image sensor, according to some embodiments. As shown in the figure, in some embodiments, pixel **100** may be formed on or in substrate **130**, e.g., a substrate made of silicon or other semiconductor materials. In some embodiments, pixel **100** may be one of a plurality of pixels of an image sensor (e.g., a CMOS image sensor, a CCD image sensor, and the like) of an image capturing device. In some embodiments, pixel **100** may include one or more circuits (e.g., a pixel readout circuit) formed using one or more transistors for reading out the image signals from pixel **100** to generate one or more analog signals (e.g., analog voltages) at a pixel output line, and/or other signal conditioning or processing circuit(s). For purposes of illustration, the circuits are not shown in FIG. 1.

(29) As shown in FIG. 1, in some embodiments, pixel **100** may include at least one photodiode comprising a charge accumulation region **102** (hereinafter “PD”), a floating diffusion region **104** (hereinafter “FD”), and at least one charge transfer region formed vertically between PD **102** and FD **104**. As shown in FIG. 1, in some embodiments, PD **102** and FD **104** may be disposed vertically one relative to another, e.g., with PD **102** placed underneath and at least partially overlapping FD **104**, and FD **104** placed proximate the top surface of pixel **100**. In addition, in some embodiments, the at least one charge transfer region may include multiple charge modulation regions, e.g., a phase 1 charge modulation region **106** (hereinafter “P1”), a phase 2 charge modulation region **108** (hereinafter “P2”), and a phase 3 charge modulation region **110** (hereinafter “P3”). As shown in FIG. 1, in some embodiments, P1 **106**, P2 **108**, and P3 **110** may be disposed vertically one on top of another between PD **102** and FD **104**. For example, P1 **106** may be formed above and at least partially overlapping PD **102**, P2 **108** may be formed above and at least partially overlapping P1 **106**, and P3 **110** may be formed (a) above and at least partially overlapping P2 **108** and (b) underneath and at least partially overlapping FD **104**. As a result, P1 **106**, P2 **108**, and P3 **110** may be stacked together and collectively form the charge transfer region vertically between PD **102** and FD **104**. In some embodiments, at least a portion of P3 **110** may optionally be disposed away and separated from FD **104** by a physical gap. For example, in some embodiments, the physical gap may be located at the top corners of P3 **110** interfacing FD **104**. The physical gap may provide a potential barrier (e.g., an electrostatic potential barrier) between P3 **110** and FD **104**, which may impede the transfer of charge from P3 **110** to FD **104**.

(30) In some embodiments, P1 **106**, P2 **108**, and P3 **110** may each be controlled by control signals (e.g., control voltages) applied to their respective control gates. Further, each control gate may include a vertical gate area (hereinafter “G”) and an associated gate contact (hereinafter “GC”). For example, the control gate of P1 **106** may include the 1.sup.st vertical gate area **112** (hereinafter “G1”), which may further be electrically connected with the 1.sup.st gate contact **122** (hereinafter “GC1”) placed within gate control area **120**. Similarly, the control gate of P2 **108** may include the 2.sup.nd vertical gate area **114** (hereinafter “G2”) and the 2.sup.nd gate contact (hereinafter “GC2”) **124** placed within gate control area **120**; the control gate of P3 **110** may include the 3.sup.rd vertical gate area **116** (hereinafter “G3”) and the 3.sup.rd gate contact **126** (hereinafter “G3”) placed within gate control area **120**. In some embodiments, G1 **112**, G2 **114**, and G3 **116** may be formed using polysilicon materials and buried within substrate **130**. In some embodiments, GC1 **122**, GC2 **124**, and GC3 **126** may be formed using metal or polysilicon materials and may serve as respective “electrodes” to receive control signals (e.g., control voltages) applied to G1 **112**, G2 **114**, and G3 **116**. As shown in FIG. 1, in some embodiments, pixel **100** may include region **132** that is isolated from substrate **130** by dielectric material **150**, at the outer layer surrounding the perimeter of pixel **100**. In some embodiments, region **132** may be formed using polysilicon or metal materials. In some embodiments, region **132** may attract holes from PD **102**, thus facilitating the accumulation of charge in PD **102** when PD **102** is exposed to light.

(31) As shown in FIG. 1, in this example, G1 **112**, G2 **114**, and G3 **116** may be disposed vertically

one on top of another (thus only G3 **116** is visible in the top view), laterally proximate their corresponding modulation regions P1 **106**, P2 **108**, and P3 **110**. In addition, G1 **112**, G2 **114**, and G3 **116** may not necessarily overlap FD **104** vertically, as shown in FIG. **1**. In this example, G1 **112**, G2 **114**, and G3 **116** may stack vertically one on top of another. In addition, G1 **112**, G2 **114**, and G3 **116** may each have a ring-shape geometry, thus completely enclosing their corresponding modulation regions P1 **106**, P2 **108**, and P3 **110**, as well as FD **102** (only laterally, as they do not overlap FD **102** vertically as described above). As a result, the potentials of P1 **106**, P2 **108**, and P3 **110** may be modulated or controlled separately by respective control signals applied to G1 **112**, G2 **114**, and G3 **116**. For example, in some embodiments, the potential of P1 **106** may be modulated by a control voltage signal applied to G1 **112**. For example, in some embodiments, P1 **106** may be implanted with one or more p-type dopants. When a positive voltage is applied to G1 **112**, the positive voltage may repel holes in the layer of P1 **106** interfacing G1 **112** away from G1 **112** to thus create a channel in the interfacing layer. As a result, charge may transfer from PD **102** through the channel of P1 **106**. Similarly, in some embodiments, the potential of P2 **108** may be modulated by a control voltage signal applied to G2 **114**, and a channel may be formed within P2 **102** for the charge to further transfer through P2 **108**. Similarly, in some embodiments, the potential of P3 **110** may be modulated by a voltage control signal applied to G3, and a channel may be formed within P3 **110** for the charge to further transfer through P3 **110**. As described in more detail below, in some embodiments, P2 **108** may serve as an in-pixel charge memory region to temporarily store charge generated from PD **102**. For example, at end of exposure, charge may be first transferred from PD **102** to P2 **108** via P1 **106**. The charge may be temporarily stored in P2 **108** until readout of pixel **100**. At the readout, the charge may then be transferred from P2 **108** to FD **104** via P3 **110**. Also, as shown in FIG. **1**, in this example, pixel **100** may be configured to receive backside illumination. But alternatively, in some embodiments, front side illumination may be implemented.

(32) In some embodiments, FD **104** may include capacitance. Thus, the transfer of charge into FD **104** may generate an analog voltage between FD **104** and the ground. The analog voltage may represent the image signals captured by pixel **100**. The analog voltage of FD **104** may further be accessed and read out of FD **104**, e.g., using a pixel readout circuit formed by one or more transistors, to generate an analog voltage at a pixel output line outside pixel **100**. In some embodiments, the image sensor may include multiple pixel output lines for reading out the FD voltages from pixels of different columns (e.g., one pixel output line for one corresponding column in the row-by-row readout). In some embodiments, the image sensor may also include one or more analog-to-digital circuits to convert the analog voltages of the pixel output lines to digital signals. In some embodiments, the image sensor may further include a transfer circuit that transmits the digital signals to an external component, e.g., an image signal processor (ISP), for further digital processing to generate the images.

(33) FIG. **2** shows example implementations for control gates of a pixel of an image sensor, according to some embodiments. As shown on the left of the figure, in some embodiments, vertical gate areas G1 **112**, G2 **114**, and G3 **116** may individually include extensions to the surface of substrate **130**. For example, G1 **112**, G2 **114**, and G3 **116** may each have an L-shape geometry, buried inside substrate **130**, which extends to the surface of substrate **130**. Further, GC1 **122**, GC2 **124**, and GC3 **126** may be attached to G1 **112**, G2 **114**, and G3 **116** at the surface of substrate **130**. Alternatively, as shown on the right of FIG. **2**, in some embodiments, G1 **112**, G2 **114**, and G3 **116** may each have a horizontal I-shape geometry, buried inside substrate **130**, whereas GC1 **122**, GC2 **124**, and GC3 **126** may extend inside substrate **130** to be electrically connected with G1 **112**, G2 **114**, and G3 **116**.

(34) FIG. **3** shows example implementations for vertical gate areas of a pixel of an image sensor, according to some embodiments. Unlike the vertical gate areas of FIG. **1**, as shown on the left of FIG. **3**, in some embodiments, the vertical gate areas G1, G2, and G3 of a pixel (e.g., the 3.sup.rd vertical gate area G3 **316** visible in the top view) may individually have a C-shape geometry, thus

only partially (not completely) enclosing FD **304** (laterally) and the corresponding charge modulation regions. Similarly, each vertical gate area G1, G2, and G3 may be electrically connected with an associated gate contact (e.g., GC1, GC2, and GC3) for receiving a respective control signal. Alternatively, as shown in the middle of FIG. **3**, in some embodiments, the vertical gate areas of a pixel (e.g., the 3rd vertical gate area G3 **356** visible in the top view) may individually have an I-shape geometry, disposed proximate FD **344** and the corresponding charge modulation regions. The cross-section view on the right of FIG. **3** represents a cross-section view for both the above described C-shape and I-shape pixels. As shown in the cross-section view, PD **342** and FD **304** (for the C-shape pixel) or **344** (for the I-shape pixel) may be still disposed vertically one relative to another. Similarly, the charge transfer region vertically between the PD and FD may include multiple charge modulation regions, e.g., P1 **346**, P2 **348**, and P3 **350** stacked vertically one on top of another. However, unlike FIG. **1**, the P1, P2, P3, and FD may have smaller cross-section areas, thus only partially rather than completely overlapping PD **342**. In some embodiments, different doping concentrations in P1, P2, and P3 of the C-shape and I-shape pixels may be introduced by implants or other methods to create sufficient capacitance and facilitate the vertical charge transfer. Note that FIGS. **1-3** are provided only as non-limiting examples for purposes of illustration. In some embodiments, the various regions of a pixel may be formed and arranged in a variety of ways. For example, in some embodiments, the different embodiments of FIGS. **1-3** may be used in combination for a given pixel, such that some of the vertical gate areas of a pixel may be formed in a ring-shape, whereas the others in an C-shape.

(35) FIGS. **4A-4D** show example timing diagrams of control signals and potential profile of a pixel of an image sensor to illustrate charge transfer, according to some embodiments. In FIG. **4A**, the horizontal axis denotes time, whereas the vertical axis denotes voltage control signals **422**, **424**, and **426** applied to gate areas G1 **412**, G2 **414**, and G3 **416** of a pixel. As shown in FIG. **4A**, operation of the pixel may include three periods **432**, **434**, and **436**. During the 1.sup.st period **432**, all the pixels (including the one shown in FIGS. **4A-4D**) of an image sensor may get exposed. For example, all the pixels may begin to be exposed to the light simultaneously at time **442**, and end the exposure simultaneously at time **444**. With the exposure, during period **432**, all the pixels may generate and accumulate charge within their respective PD regions. FIG. **4B** represents a potential profile for the different regions of a pixel of the image sensor corresponding to period **432**. As shown in FIG. **4B**, during period **432**, no active control voltages **422**, **424**, and **426** are applied to G1 **412**, G2 **414**, and G3 **416** (or applied by a negative voltage if needed). For example, G1 **412**, G2 **414**, and G3 **416** may be biased at zero voltage (or a negative voltage if needed). Thus, it shows that the charge may be accumulated inside PD **402**, and P1 **406**, P2 **408**, and P3 **410** may be turned off. In some embodiments, the different regions of the pixel may have different doping types and/or doping concentrations. For example, in some embodiments, PD **402** may be a n-type region formed with one or more n-type dopants, P1 **406**, P2 **408**, and P3 **410** may be p-type regions formed with one or more p-type dopants, P1, P2, P3 might also be n-type regions with different doping concentrations. and FD **404** may be a n-type region formed with one or more n-type dopants. Thus, as shown in FIG. **4B**, the potential profile of PD **402**, P1 **406**, P2 **408**, P3 **410**, and FD **404** may have a multi-stepped shape. In addition, in this example, P1 **406**, P2 **408**, and P3 **410** may also have the same doping concentration. Thus, in FIG. **4B**, at turn-off, P1 **406**, P2 **408**, and P3 **410** may have the same potential. Alternatively, in some embodiments, P1 **406**, P2 **408**, and P3 **410** may have different doping concentration, which may cause different potentials (e.g., in a multi-stepped shape) between these regions.

(36) During the 2.sup.nd period **434**, after the exposure ends for all the pixels of the image sensor, all the pixels may transfer the charge from their PD regions to their in-pixel memory regions, e.g., their P2 regions, simultaneously at the same time. For example, as shown in FIG. **4A**, active control voltages **422** and **424** may be applied to G1 **412** and G2 **414**, whereas no active control voltage **426** may be applied to G3 **410**. In other words, P1 **406** and P2 **408** may be turned on,



whereas P3 410 may stay off. For example, as shown in FIG. 4A, positive voltages may be applied to G1 412 and G2 414, whereas G3 416 may be still biased at the zero voltage (or negative voltage if needed). As described above, in some embodiments, the positive voltages may repel holes within the interfacing layers of P1 406 and P2 408 away from P1 406 and P2 408, thus allowing at least some of the charge to be transferred from PD 402 into P1 406 and P2 408. In addition, as described above, at turn-off, P1 406 and P2 408 may have the same potential. Thus, in some embodiments, voltages of different values may be applied to G1 412 and G2 414, thus causing the stepped-potential profile between P1 406 and P2 408, as shown in FIG. 4C corresponding to period 434. As a result, the charge may be transferred from PD 402 to P2 408 via P1 406, as shown in FIG. 4C. As described above, in some embodiments, P1 406 and P2 408 may have different doping concentrations and thus different potentials at turn-off. Thus, in that case, voltages of the same value may be applied to G1 412 and G2 414 to still maintain the stepped-potential profile between P1 406 and P2 408 shown in FIG. 4C for the charge to be transferred from PD 402 to P2 408 via P1 406.

(37) As described, once the charge is transferred to P2 408, P2 408 may function as an in-pixel charge memory region to temporarily store the charge there until readout of the pixel. Again, as described above, the pixels of the image sensor may be read out individually at respective times. For example, in a row-by-row readout mode, pixels of the same row may be read out at the same time, whereas pixels in the same column but different rows may be read out sequentially one by one. As shown in FIG. 4A, at time 446 the pixel may be selected to be read out. During the 3.sup.rd period 436, active control voltages 424 and 426 may be applied to G2 414 and G3 416, whereas no active control voltage 422 may be applied to G1 412. In other words, P2 408 and P3 410 may be turned on, whereas P1 406 may be turned off. For example, as shown in FIG. 4A, positive voltages may be applied to G2 414 and G3 416, whereas G1 412 may be biased at the zero voltage (or negative voltage if needed). Similar to what is described above, voltages of different values may be applied to G2 414 and G3 416, thus creating the stepped-potential profile between P2 408 and P3 410 to transfer the charge from P2 408 to FD 404 via P3 410, as shown in FIG. 4D corresponding to period 436. Alternatively, P2 408 and P3 410 may have different doping concentrations, and voltages of the same value may be applied to G2 414 and G3 416, to still maintain the stepped-potential profile between P2 408 and P3 410 to transfer the charge from P2 408 to FD 404 via P3 410. As shown in FIG. 4A, in some embodiments, active control voltage 424 may be continuously applied to G2 414 during periods 434 and 436. Alternatively, in some embodiments, active control voltage 424 may be removed at the end of period 436 when the charge has been transferred from PD 402 to P2 408, and then re-applied at or near the beginning of period 436 to transfer the charge from P2 408 to FD 404, as shown by the dashed line of FIG. 4A. In combination of FIGS. 4A-4D, using appropriate control signals, an image sensor may be controlled to operate in the global shutter mode. All pixels of the image sensor may get exposed and transfer accumulated charge from their PD regions to in-pixel memory regions, e.g., their P2 regions, simultaneously during the same time interval. Then the pixels may be individually read out, e.g., during their respective rolling readout intervals, in which the charge may be transferred from the individual pixels' P2 regions to their FD regions.

(38) FIG. 5A shows a cross-section view of an example pixel and a block diagram of an image sensor, including the readout and image signal processing circuits, according to some embodiments. As shown in FIG. 5A, pixel 500 may be one of a plurality of pixels of image sensor 550. Also, as described above, in some embodiments, the plurality of pixels may be organized as a pixel array having one or more rows and one or more columns of pixels. Similar to the pixels described above, pixel 500 may include at least one PD, a FD, and a vertical charge transfer region between the PD and FD, which includes P1, P2, and P3. In addition, for purposes of illustration, FIG. 5A shows at least some of the transistors that may be used to form a pixel readout circuit for reading out the image signals of pixel 500. For example, in some embodiments, the pixel readout

circuit may include reset transistor (“RG”) 542, source-follower transistors (“SF”) 544, and read selection transistor (“RS”) 546. As shown in FIG. 5A, the FD may be coupled to a reset voltage VDD via RG 542. In some embodiments, RG 542 may be selectively turned on to reset the voltage of the FD to VDD. Further, as shown in FIG. 5A, the FD may be also coupled with SF 544 and one or more RS 546. In some embodiments, SF 544 and RS 546 may be turned on to couple the FD to the pixel output line, through which the voltage of the FD may be accessed and read out. In some embodiments, SF 544 may provide a voltage buffer for the voltage of the FD, whereas RS 546 may be selectively turned on to couple the FD with the pixel output line for reading out the voltage of the FD. During readout, SF 544 and RS 546 may be first turned on to couple the FD to the pixel output line. Next, RG 542 may be turned on to reset the voltage of the FD to the reset voltage VDD. The voltage of the FD may be sampled, e.g., using amplifier and analog-to-digital circuits 554, as the 1.<sup>sup</sup>.st sample of the voltage of the FD. Next, RG 542 may be turned off, and the charge transfer region between the PD and FD may be turned on to transfer charge from the PD to the FD, as described above in FIGS. 1-4. As described above, transfer of the charge may generate an analog voltage across the capacitance C of the FD. The voltage of the FD may again be accessed and read out at the pixel output line through SF 544 and RS 546, as described above. The voltage of the FD may be sampled, e.g., using amplifier and analog-to-digital circuits 554, as the 2.<sup>sup</sup>.nd sample of the voltage of the FD. The difference between the first sample and second sample may be calculated to cancel out the reset voltage VDD, and the differential voltage may be determined as the final image signal from pixel 500. The image signal may be further processed, e.g., using digital processing circuit(s) 556. The image signal may be transmitted, e.g., using data transfer I/O circuit(s) 558, to an image signal processor (ISP) to be processed to produce an image. In addition, as shown in FIG. 5A, image sensor 550 may also include row logic and drivers circuit(s) 552 and global logic and clocking circuit(s) 560 in order to generate the appropriate control signals for the plurality of pixels of image sensor 550.

(39) FIG. 5B shows example control signals applied to different components of a plurality of pixels of image sensor 550, according to some embodiments. FIG. 5B shows information similar to what is described above with respect to FIGS. 4A-4D, but for a plurality of pixels instead of one individual pixel. For example, as shown in FIG. 5B, the control signals shown here include control signals applied to the pixels of different rows, e.g., row 0, row 1, row 2, . . . , row n. In this example, image sensor 550 may operate in a global shutter mode. For example, all the pixels may be exposed during a 1.<sup>sup</sup>.st period, e.g., an exposure period (similar to period 432 of FIGS. 4A-4D). In addition, all the pixels may transfer the charge for their PD regions to their in-pixel memory regions, e.g., their P2 regions, simultaneously during a 2.<sup>sup</sup>.nd period, e.g., a global shutting period (similar to period 434). Further, the pixels of the image sensor may be read out individually at respective times during a 3rd period, e.g., rolling readout period (similar to period 436). Similar to what is described above in FIGS. 4A-4B, as shown in FIG. 5B, during the 1.<sup>sup</sup>.st period, no active control voltages may be applied to the control gates of the pixels of image sensor 550, and thus all the gates may be turned off, so that the pixels may accumulate charge in their respective PD regions. In addition, similar to what is described above in FIGS. 4A and 4C, as shown in FIG. 5B, during the 2.<sup>sup</sup>.nd period, active control voltages may be applied to the control gates G1 and G2 of the pixels of image sensor 550, and thus all the pixels may transfer the charge from their respective PD regions to P2 regions via P1 regions simultaneously. Moreover, similar to what is described above in FIGS. 4A and 4D, as shown in FIG. 5B, during the 3.<sup>sup</sup>.rd period, active control voltages may be applied to the control gates G2 and G3 of the pixels of image sensor 550 at respective readout times, and thus the respective pixels may transfer the charge from their respective P2 regions to FD regions via the P2 regions. Also, as described above in FIG. 5A, different control signals may be applied to the transistors (e.g., the RST and RS transistors) of the respective pixels to select and read out the respective pixels, as shown in FIG. 5B. Also, as shown in FIG. 5B, these transistor control signals may be applied at different times for pixels of different

rows to represent the rolling readout operation. For example, the control signals to RSG[0] and RS[0] of a pixel of row 0 may be applied ahead of the control signal to RSG[1] and RS[1] of a pixel of row 1, meaning that the pixel of row 0 may be read before the pixel of row 1. As described above, the charge may be transferred from PD regions to P2 regions for all the pixels globally at the same time during the 2.sup.nd period. However, the pixels of the different rows may be read out at different times. Thus, the P2 regions may thus serve as in-pixel charge memory regions to temporarily store the charge for each individual pixel.

(40) FIG. 6 show top view and cross-section view of another example pixel of an image sensor, according to some embodiments. As shown in FIG. 6, in some embodiments, the gate areas G1 612, G2, 614, and G3 616 of pixel 600 may be placed in a hexogen-shape arrangement. In addition, G1 612, G2, 614, and G3 616 may be each electrically connected with a gate contact, e.g., GC1 622, GC2 624, and GFC3 626. Also, unlike the pixels described above, in FIG. 6, GC1 622, GC2 624, and GFC3 626 may not necessarily overlap each other. Instead, they may be placed around the perimeter of the hexogen shape, where the gate area for each charge modulation region may include a first portion of the gate area connected with a first portion of a gate contact and a second portion of the gate area connected with a second portion of the gate contact, each pair located at one of two opposite sides of the corresponding charge modulation region. For example, as shown in the cross-section view along AA', G1 612 of P1 606 may include a first G1 portion on the left and a second G1 portion on the right of P1 606. The left G1 portion may be electrically connected with a first GC portion 622, whereas the right G1 portion may be electrically connected with a second GC portion 622, and the first and second GC portions may be electrically connected with each other. In addition, as shown in FIG. 6, in some embodiments, G1 612, G2 614, and G3 616 may be buried inside substrate 630 at different depths. For example, G1 612 may be buried at the deepest depth closer to PD 602, G3 616 may be buried at the shallowest depth closer to the surface of substrate 630, whereas G2 614 may be buried in the middle.

(41) FIG. 7 shows example isolation structures of a pixel of an image sensor, according to some embodiments. The figure on the left shows a partial deep trench isolation (DTI) for a pixel, according to some embodiments. As shown in the figure, in some embodiments, the DTI may include region 732 formed using polysilicon or metal materials. In some embodiments, region 732 may be formed within a trench (filled with the polysilicon or metal materials) that is isolated from substrate 730 by dielectric material 750, and the trench may enclose the PD region of the pixel. In some embodiments, region 732 may be biased to create passivation for the trench surface to isolate the pixel from other pixels next to the pixel. Alternatively, in some embodiments, the trench may be filled with high-k dielectric material(s). In this case, region 732 may not necessarily need to be biased. The figure on the right shows an alternative isolation structure, a deep doping well isolation, for a pixel. As shown in this figure, the trench for region 752 may extend all the way to the surface of the pixel. The trench may be implanted with dopants to create electrical isolation between pixels.

(42) FIG. 8 shows an example split pixel for autofocus application, according to some embodiments. As shown in FIG. 8, pixel 800 may include at least two PDs, left PD 802 and right PD 862. In some embodiments, PD 802 and PD 862 may be isolated from each other using one or more isolation structures. For example, as shown in FIG. 8, in some embodiments, the two PDs may be isolated by partial deep trench insulation (DTI) region 832, as described above, located between PD 802 and PD 862. In some embodiments, each PD may be associated with a FD and vertical charge transfer region including P1, P2, and P3. For example, left PD 802 may be associated with left P1 (L-P1), left P2 (L-P2), left P3 (L-P3), and left FD (L-FD), whereas right PD 862 may be associated with right P1 (R-P1), right P2 (R-P2), right P3 (R-P3), and right FD (R-FD), similar to what is described above in FIGS. 1-7. Further, L-FD and R-FD may be electrically connected together, such that the image signals of L-FD and R-FD may be read out to a pixel output line using the same pixel readout circuit. During operation, gate areas LG1 (for L-P1) and

RG1 (for R-P1) may receive control signals at or around the same time, and LG2 (for L-P2) and RG2 (R-P2) may receive the control signals at or around the same time, such that the charge may be transferred from PD **802** to L-FD simultaneously as the charge transfer from PD **862** to R-FD. However, gate areas LG3 (for L-P3) and RG3 (for R-P3) may receive control signals at different times so that the charge may be transferred from L-P2 to L-FD and from R-P3 to R-FD at different times. In other words, the images signals for L-PD **802** and R-PD **862** of pixel **800** may be read out sequentially. In some embodiments, pixel **800** may be used for autofocus of an image capturing device. For example, when light comes at an angle to pixel **800**, the amount of light captured by L-PD **802** and R-PD **862** may be different. Accordingly, the two PDs may accumulate different amounts of charge and thus cause different output voltages at readout. In some embodiments, the difference between the output voltages from L-PD **802** and R-PD **862** may be used to adjust and/or perform autofocus of the image capturing device.

(43) FIG. **9** shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments. As shown in FIG. **9**, in some embodiments, the vertical gates and DTI isolation of a pixel may be formed in the same trench. For example, in some embodiments, gate areas of pixel **900**, G1 **912**, G2 **914**, and G3 **916**, may be formed in the same trench **930** inside the substrate of pixel **900**, as shown especially in the cross-sections of FIG. **9**. G1 **912**, G2 **914**, and G3 **916** may each receive a control signal (e.g., a control voltage) through an associated gate contact, e.g., GC1 **922**, GC2 **924**, and GC3, **926** respectively. Similar to what is described above in FIGS. **1-8**, pixel **900** may include PD **902**, FD **904**, and a vertical charge transfer region between PD **902** and FD **904**, and the vertical charge region may include multiple charge modulation regions P1 **906**, P2 **908**, and P3 **910**. In some embodiments, pixel **900** may be operated in a global shutter mode similarly as the above described pixels, where P2 **908** may function as an in-pixel charge memory region for pixel **900**.

(44) FIG. **10** shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments. As shown in FIG. **10**, the structure of pixel **1000** may be similar to that of pixel **900**. However, the P1, P2, P3, and FD regions of pixel **1000** may have smaller cross-section areas such that these regions only partially rather than completely overlapping the PD underneath PD region. In some embodiments, different doping concentrations in P1, P2, and P3 may be introduced by implants or other methods to create sufficient capacitance and facilitate the vertical charge transfer. As shown in FIG. **10**, in some embodiments, there may be a set of gate areas for a neighbored pixel (not shown) on the right of pixel **1000**, which are disposed in the trench proximate pixel **1000**. Thus, in some embodiments, there may be isolation region **1032** between the gate areas (G1, G2, and G3) of pixel **1000** and the gate areas of the other pixel. In some embodiments, isolation region **1032** may be implanted with dopants to provide the isolation.

(45) FIG. **11** shows top view and cross-section view of another example pixel of an image sensor, according to some embodiments. In this example, FD **1104**, P1 **1106**, P2 **1108**, and P3 **1110** may be moved to the corner of a pixel (e.g., for PD **1102**), e.g., to the right corner of the pixel, as shown in the cross-section view of FIG. **11**. In some embodiments, one performance parameter of a global shutter image sensor is parasitic light sensitivity (PLS), which quantifies the sensor sensitivity to light when the shutter is supposed closed. Thus, the image signals generated in this case may be considered the background noise. Thus, in some embodiments, it may be desired to minimize or at least reduce the value of PLS. By moving P1 **1106**, P2 **1108**, and P3 **1110** to the corner of a pixel, DTI region **1132** may be used as the shield of incident light to reduce the amount of light going into P2 **1108** to thus improve the PLS performance. In addition, in some embodiments, the pixel may include doping region **1152** to provide a potential barrier to prevent the charge from PD **1102** from being transferred to the P1, P2, P3, and FD at the left corner of the pixel (since those regions are for the pixel on the left). Instead, the charge of PD **1102** may be guided to be transferred correctly to FD **1104** via P1 **1106**, P2 **1108**, and P3 **1110**, as these regions correspond to PD **1102**, as indicated by the left arrow in FIG. **11**. Similarly, for the pixel on the right (e.g., for PD **1162**), the pixel may

also include doping region **1172** to block the charge from PD **1162** from being erroneously transferred from PD **1162** to P1 **1106**, P2 **1108**, P3 **1110**, and FD **1104**. Instead, the charge of PD **1162** may be transferred to the FD via P1, P2, and P3 at the right corner of the right pixel, as indicated by the right arrow in FIG. **11**. In some embodiments, the doping type of doping regions **1152** and **1172** may be opposite to the doping type of PD **1102** and PD **1162**.

(46) FIG. **12** is a block diagram of an example image capturing device, according to some embodiments. As shown in FIG. **12**, in some embodiments, image capturing device **1200** may include one or more lenses **1202** and image sensor **1204**. In some embodiments, image capturing device **1200** may capture light from an environment, and the light may pass through lenses **1202** to reach image sensor **1204**. In some embodiments, image sensor **1204** may include a plurality of pixels similar to the pixels described above, where each may include an in-pixel charge memory region for a global shutter operation. Also, in some embodiments, image capturing device **1200** may include infrared cutoff filter (IRCF) **1206** placed between lenses **1202** and image sensor **1204** to block infrared light from reaching image sensor **1204**. As shown in FIG. **12**, in this example, image sensor **1204** and IRCF **1206** may be mounted on substrate **1208**, and image sensor **1204** may be placed upside down so as to receive backside illumination. But as described above, alternatively, in some embodiments, front side illumination may be implemented on image sensor **1204** having the above described pixels. In some embodiments, image sensor **204** may be a CMOS image sensor.

(47) FIG. **13** is a flowchart showing an example method for using an in-pixel charge memory region to operate an image sensor in a global shutter mode, according to some embodiments. As shown in FIG. **13**, in some embodiments, an image sensor including a plurality of pixels may be provided, as indicated by block **1302**. In some embodiments, each pixel of the image sensor may include (a) a photodiode comprising a charge accumulation region (PD), (b) a floating diffusion region (FD), and (c) a gate transfer region vertically between the PD and FD. In addition, in some embodiments, the vertical gate transfer region may further include (i) a plurality of charge modulation regions (P1, P2, and P3) formed vertically between the PD and FD; and (ii) control gates for the respective charge modulation regions. As shown in FIG. **13**, in some embodiments, for a particular one of the pixels, the pixel may accumulate charge in its PD when exposed to light during a 1.sup.st period, as indicated by block **1304**. As described above, in some embodiments, the image sensor may operate in the global shutter mode, and thus all the pixels including the particular pixel may begin exposure and end exposure simultaneously during the first period. As shown in FIG. **13**, in some embodiments, for the particular pixel, at least some of the charge of the PD may be transferred from the PD to P2 via P1 during a 2.sup.nd period, as indicated by block **1306**. As described above, in some embodiments, the transfer of charge from PD to P2 may be a global operation for all the pixels of the image sensor. In other words, all the pixels including the particular pixel may transfer their charge from PD to P2 simultaneously during the 2.sup.nd period. In addition, as described above, in some embodiments, P2 may serve as an in-pixel charge memory region for the particular pixel. Thus, once the charge is transferred to P2, it may be temporarily stored there until the readout of the particular pixel. As shown in FIG. **13**, in some embodiments, the charge transferred to P2 may further be transferred from P2 to FD via P1 during a 3.sup.rd period, as indicated by block **1308**. As described above, even in the global shutter mode, the individual pixels of the image sensor may not necessarily be read out simultaneously. Instead, the pixels may be read out line by line in sequence. Thus, the 3.sup.rd period to transfer the charge from P2 to FD for the particular pixel may or may not be the same for the readout period of another pixel.

(48) FIG. **14** illustrates a schematic representation of an example device **1400** that may include an image capturing device (e.g., a camera) having an image sensor that includes pixels with the above described in-pixel charge memory regions, according to some embodiments. In some embodiments, the device **1400** may be a mobile device and/or a multifunction device. In various embodiments,

the device **1400** may be any of various types of devices, including, but not limited to, a personal computer system, desktop computer, laptop, notebook, tablet, slate, pad, or netbook computer, mainframe computer system, handheld computer, workstation, network computer, a camera, a set top box, a mobile device, an augmented reality (AR) and/or virtual reality (VR) headset, a consumer device, video game console, handheld video game device, application server, storage device, a television, a video recording device, a peripheral device such as a switch, modem, router, or in general any type of computing or electronic device.

(49) In some embodiments, the device **1400** may include a display system **1402** (e.g., comprising a display and/or a touch-sensitive surface) and/or one or more cameras **1404**. In some non-limiting embodiments, the display system **1402** and/or one or more front-facing cameras **1404a** may be provided at a front side of the device **1400**, e.g., as indicated in FIG. **14**. Additionally, or alternatively, one or more rear-facing cameras **1404b** may be provided at a rear side of the device **1400**. In some embodiments comprising multiple cameras **1404**, some or all of the cameras may be the same as, or similar to, each other. Additionally, or alternatively, some or all of the cameras may be different from each other. In various embodiments, the location(s) and/or arrangement(s) of the camera(s) **1404** may be different than those indicated in FIG. **14**.

(50) Among other things, the device **1400** may include memory **1406** (e.g., comprising an operating system **1408** and/or application(s)/program instructions **1410**), one or more processors and/or controllers **1412** (e.g., comprising CPU(s), memory controller(s), display controller(s), and/or camera controller(s), etc.), and/or one or more sensors **1416** (e.g., orientation sensor(s), proximity sensor(s), and/or position sensor(s), etc.). In some embodiments, the device **1400** may communicate with one or more other devices and/or services, such as computing device(s) **1418**, cloud service(s) **1420**, etc., via one or more networks **1422**. For example, the device **1400** may include a network interface (e.g., network interface **810**) that enables the device **1400** to transmit data to, and receive data from, the network(s) **1422**. Additionally, or alternatively, the device **1400** may be capable of communicating with other devices via wireless communication using any of a variety of communications standards, protocols, and/or technologies.

(51) FIG. **15** illustrates a schematic block diagram of an example computing device, referred to as computer system **1500**, that may include or host embodiments of an image capturing device (e.g., a camera) having an image sensor that includes pixels with the above described in-pixel charge memory regions, according to some embodiments. In addition, computer system **1500** may implement methods for controlling operations of the camera and/or for performing image processing images captured with the camera. In some embodiments, the device **1400** (described herein with reference to FIG. **14**) may additionally, or alternatively, include some or all of the functional components of the computer system **1500** described herein.

(52) The computer system **1500** may be configured to execute any or all of the embodiments described above. In different embodiments, computer system **1500** may be any of various types of devices, including, but not limited to, a personal computer system, desktop computer, laptop, notebook, tablet, slate, pad, or netbook computer, mainframe computer system, handheld computer, workstation, network computer, a camera, a set top box, a mobile device, an augmented reality (AR) and/or virtual reality (VR) headset, a consumer device, video game console, handheld video game device, application server, storage device, a television, a video recording device, a peripheral device such as a switch, modem, router, or in general any type of computing or electronic device.

(53) In the illustrated embodiment, computer system **1500** includes one or more processors **1502** coupled to a system memory **1504** via an input/output (I/O) interface **1506**. Computer system **1500** further includes one or more cameras **1508** coupled to the I/O interface **1506**. Computer system **1500** further includes a network interface **1510** coupled to I/O interface **1506**, and one or more input/output devices **1512**, such as cursor control device **1514**, keyboard **1516**, and display(s) **1518**. In some cases, it is contemplated that embodiments may be implemented using a single instance of computer system **1500**, while in other embodiments multiple such systems, or multiple nodes

making up computer system **1500**, may be configured to host different portions or instances of embodiments. For example, in one embodiment some elements may be implemented via one or more nodes of computer system **1500** that are distinct from those nodes implementing other elements.

(54) In various embodiments, computer system **1500** may be a uniprocessor system including one processor **1502**, or a multiprocessor system including several processors **1502** (e.g., two, four, eight, or another suitable number). Processors **1502** may be any suitable processor capable of executing instructions. For example, in various embodiments processors **1502** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. Also, in some embodiments, one or more of processors **1502** may include additional types of processors, such as graphics processing units (GPUs), application specific integrated circuits (ASICs), etc. In multiprocessor systems, each of processors **1502** may commonly, but not necessarily, implement the same ISA. In some embodiments, computer system **1500** may be implemented as a system on a chip (SoC). For example, in some embodiments, processors **1502**, memory **1504**, I/O interface **1506** (e.g. a fabric), etc. may be implemented in a single SoC comprising multiple components integrated into a single chip. For example, an SoC may include multiple CPU cores, a multi-core GPU, a multi-core neural engine, cache, one or more memories, etc. integrated into a single chip. In some embodiments, an SoC embodiment may implement a reduced instruction set computing (RISC) architecture, or any other suitable architecture.

(55) System memory **1504** may be configured to store program instructions **1520** accessible by processor **1502**. In various embodiments, system memory **1504** may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. Additionally, existing camera control data **1522** of memory **1504** may include any of the information or data structures to implement the techniques described above. In some embodiments, program instructions **1520** and/or data **1522** may be received, sent or stored upon different types of computer-accessible media or on similar media separate from system memory **1504** or computer system **1500**. In various embodiments, some or all of the functionality described herein may be implemented via such a computer system **1500**.

(56) In one embodiment, I/O interface **1506** may be configured to coordinate I/O traffic between processor **1502**, system memory **1504**, and any peripheral devices in the device, including network interface **1510** or other peripheral interfaces, such as input/output devices **1512**. In some embodiments, I/O interface **1506** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **1504**) into a format suitable for use by another component (e.g., processor **1502**). In some embodiments, I/O interface **1506** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **1506** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **1506**, such as an interface to system memory **1504**, may be incorporated directly into processor **1502**.

(57) Network interface **1510** may be configured to allow data to be exchanged between computer system **1500** and other devices attached to a network **1524** (e.g., carrier or agent devices) or between nodes of computer system **1500**. Network **1524** may in various embodiments include one or more networks including but not limited to Local Area Networks (LANs) (e.g., an Ethernet or corporate network), Wide Area Networks (WANs) (e.g., the Internet), wireless data networks, some other electronic data network, or some combination thereof. In various embodiments, network interface **1510** may support communication via wired or wireless general data networks, such as any suitable type of Ethernet network, for example; via telecommunications/telephony networks

such as analog voice networks or digital fiber communications networks; via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

(58) Input/output devices **1512** may, in some embodiments, include one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or accessing data by one or more computer systems **1500**. Multiple input/output devices **1512** may be present in computer system **1500** or may be distributed on various nodes of computer system **1500**. In some embodiments, similar input/output devices may be separate from computer system **1500** and may interact with one or more nodes of computer system **1500** through a wired or wireless connection, such as over network interface **1510**.

(59) Those skilled in the art will appreciate that computer system **1500** is merely illustrative and is not intended to limit the scope of embodiments. In particular, the computer system and devices may include any combination of hardware or software that can perform the indicated functions, including computers, network devices, Internet appliances, PDAs, wireless phones, pagers, etc. Computer system **1500** may also be connected to other devices that are not illustrated, or instead may operate as a stand-alone system. In addition, the functionality provided by the illustrated components may in some embodiments be combined in fewer components or distributed in additional components. Similarly, in some embodiments, the functionality of some of the illustrated components may not be provided and/or other additional functionality may be available.

(60) Those skilled in the art will also appreciate that, while various items are illustrated as being stored in memory or on storage while being used, these items or portions of them may be transferred between memory and other storage devices for purposes of memory management and data integrity. Alternatively, in other embodiments some or all of the software components may execute in memory on another device and communicate with the illustrated computer system via inter-computer communication. Some or all of the system components or data structures may also be stored (e.g., as instructions or structured data) on a computer-accessible medium or a portable article to be read by an appropriate drive, various examples of which are described above. In some embodiments, instructions stored on a computer-accessible medium separate from computer system **1500** may be transmitted to computer system **1500** via transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link. Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include a non-transitory, computer-readable storage medium or memory medium such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc. In some embodiments, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

(61) The methods described herein may be implemented in software, hardware, or a combination thereof, in different embodiments. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, etc. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various embodiments described herein are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described herein as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component. These and



other variations, modifications, additions, and improvements may fall within the scope of embodiments as defined in the claims that follow.

## Claims

1. An image sensor, comprising: a plurality of pixels, wherein each of the pixels comprises: at least one photodiode comprising a charge accumulation region; at least one floating diffusion region; and at least one charge transfer region formed between the charge accumulation region and floating diffusion region, wherein the at least one charge transfer region is vertically below the at least one floating diffusion region, wherein the charge transfer region includes a first charge modulation region, a second charge modulation region, and a third charge modulation region, and wherein the pixel is controlled to: accumulate charge in the charge accumulation region based on light exposure; transfer at least some of the charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.
2. The image sensor of claim 1, wherein: the first charge modulation region is formed above and at least partially overlapping the charge accumulation region of the photodiode; the second charge modulation region formed above and at least partially overlapping the first charge modulation region; the third charge modulation region formed above and at least partially overlapping the second charge modulation region; and the floating diffusion region formed above and at least partially overlapping the third charge modulation region.
3. The image sensor of claim 1, wherein: the first and second charge modulation regions are turned on to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and the second and third charge modulation regions are turned on to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.
4. The image sensor of claim 3, wherein: the first, second, and third charge modulation regions are turned on responsive to respective positive voltages applied to respective gate control areas of the first, second, and third charge modulation regions; and the first, second, and third charge modulation regions are turned off responsive to removal of the respective positive voltages from respective gate control areas of the first, second, and third charge modulation regions.
5. The image sensor of claim 4, wherein: the first, second, and third charge modulation regions have a dopant of a same concentration, such that: respective positive voltages of different values are applied to the respective gate control areas of the first and second charge modulation regions to create a stepped-potential profile between the first and second charge modulation regions to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and respective positive voltages of different values are applied to the respective gate control areas of the second and third charge modulation regions to create a stepped-potential profile between the second and third charge modulation regions to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.
6. The image sensor of claim 4, wherein: the first, second, and third charge modulation regions have a dopant of different concentrations so as to create stepped-potential profiles between the first, second, and third charge modulation regions, such that: respective positive voltages of a same value are applied to the respective gate control areas of the first and second charge modulation regions to maintain the stepped-potential profile between the first and second charge modulation regions to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and respective positive voltages of a same value are applied to the respective gate control areas of the second and third charge

modulation regions to maintain the stepped-potential profile between the second and third charge modulation regions to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.

7. The image sensor of claim 4, wherein the respective gate control areas of the first, second, and third charge modulation regions are formed in a shape to at least partially encircle a lateral perimeter of the floating diffusion region.

8. The image sensor of claim 1, wherein: the plurality of pixels is organized as a pixel array to divide the pixels into multiple rows and multiple columns; and the plurality of pixels is controlled to operate in a global shutter mode, such that: pixels of the multiple rows and multiple columns accumulate charge in their respective charge accumulation regions based on light exposure simultaneously; pixels of the multiple rows and multiple columns transfer at least some of the charge from their respective charge accumulation regions to their respective second charge modulation regions via their respective first charge modulation regions simultaneously; pixels of same rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions simultaneously; and pixels of different rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions sequentially in order of the rows.

9. The image sensor of claim 1, wherein the charge accumulation region is a n-type region, the first, second, and third charge modulation regions are p-type regions, and the floating diffusion region is a n-type region.

10. The image sensor of claim 1, wherein the pixel further comprises pixel readout circuitry including at least one of: a reset switch for resetting a voltage of the floating diffusion region to a reset voltage, a source follower switch for buffering the voltage of the floating diffusion region, or a pixel selection switch for selectively coupling the floating diffusion region to a pixel output line for reading out the voltage of the floating diffusion region.

11. A system, comprising: one or more lenses; an image sensor comprising: a plurality of pixels configured to receive light through the lenses, wherein each of the pixels comprises: at least one photodiode comprising a charge accumulation region; at least one floating diffusion region; and at least one charge transfer region formed between the charge accumulation region and floating diffusion region, wherein the at least one charge transfer region is vertically below the at least one floating diffusion region, wherein the charge transfer region includes a first charge modulation region, a second charge modulation region, and a third charge modulation region, and wherein the pixel is controlled to: accumulate charge in the charge accumulation region based on light exposure; transfer at least some of the charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region to generate an image signal at the floating diffusion region; and circuitry to read out image signals from the plurality of pixels.

12. The system of claim 11, wherein: the first charge modulation region is formed above and at least partially overlapping the charge accumulation region of the photodiode; the second charge modulation region formed above and at least partially overlapping the first charge modulation region; the third charge modulation region formed above and at least partially overlapping the second charge modulation region; and the floating diffusion region formed above and at least partially overlapping the third charge modulation region.

13. The system of claim 11, wherein: the first and second charge modulation regions are turned on to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and the second and third charge modulation regions are turned on to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.

14. The system of claim 13, wherein: the first, second, and third charge modulation regions are turned on responsive to respective positive voltages applied to respective gate control areas of the first, second, and third charge modulation regions; and the first, second, and third charge modulation regions are turned off responsive to removal of the respective positive voltages from respective gate control areas of the first, second, and third charge modulation regions.

15. The system of claim 14, wherein: the first, second, and third charge modulation regions have a dopant of a same concentration, such that: respective positive voltages of different values are applied to the respective gate control areas of the first and second charge modulation regions to create a stepped-potential profile between the first and second charge modulation regions to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and respective positive voltages of different values are applied to the respective gate control areas of the second and third charge modulation regions to create a stepped-potential profile between the second and third charge modulation regions to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.

16. The system of claim 14, wherein: the first, second, and third charge modulation regions have a dopant of different concentrations so as to create stepped-potential profiles between the first, second, and third charge modulation regions, such that: respective positive voltages of a same value are applied to the respective gate control areas of the first and second charge modulation regions to maintain the stepped-potential profile between the first and second charge modulation regions to transfer the at least some charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and respective positive voltages of a same value are applied to the respective gate control areas of the second and third charge modulation regions to maintain the stepped-potential profile between the second and third charge modulation regions to transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region.

17. The system of claim 11, wherein: the plurality of pixels is organized as a pixel array to divide the pixels into multiple rows and multiple columns; and the plurality of pixels is controlled to operate in a global shutter mode, such that: pixels of the multiple rows and multiple columns accumulate charge in their respective charge accumulation regions based on light exposure simultaneously; pixels of the multiple rows and multiple columns transfer at least some of the charge from their respective charge accumulation regions to their respective second charge modulation regions via their respective first charge modulation regions simultaneously; pixels of same rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions simultaneously; and pixels of different rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions sequentially in order of the rows.

18. A device, comprising: an image capturing device, comprising: one or more lenses; an image sensor comprising: a plurality of pixels configured to receive light through the lenses, wherein each of the pixels comprises: at least one photodiode comprising a charge accumulation region; at least one floating diffusion region; and at least one charge transfer region formed between the charge accumulation region and floating diffusion region, wherein the at least one charge transfer region is vertically below the at least one floating diffusion region, wherein the charge transfer region includes a first charge modulation region, a second charge modulation region, and a third charge modulation region, and wherein the pixel is controlled to: accumulate charge in the charge accumulation region based on light exposure; transfer at least some of the charge from the charge accumulation region to the second charge modulation region via the first charge modulation region; and transfer the at least some charge from the second charge modulation region to the floating diffusion region via the third charge modulation region to generate an image signal at the floating

diffusion region; and circuitry to read out image signals from the plurality of pixels; and an image signal processor configured to process the image signals to produce one or more images.

19. The device of claim 18, wherein: the first charge modulation region is formed above and at least partially overlapping the charge accumulation region of the photodiode; the second charge modulation region formed above and at least partially overlapping the first charge modulation region; the third charge modulation region formed above and at least partially overlapping the second charge modulation region; and the floating diffusion region formed above and at least partially overlapping the third charge modulation region.

20. The device of claim 18, wherein: the plurality of pixels is organized as a pixel array to divide the pixels into multiple rows and multiple columns; and the plurality of pixels is controlled to operate in a global shutter mode, such that: pixels of the multiple rows and multiple columns accumulate charge in their respective charge accumulation regions based on light exposure simultaneously; pixels of the multiple rows and multiple columns transfer at least some of the charge from their respective charge accumulation regions to their respective second charge modulation regions via their respective first charge modulation regions simultaneously; pixels of same rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions simultaneously; and pixels of different rows transfer the at least some charge from their respective second charge modulation regions to their respective floating diffusion regions via their respective third charge modulation regions sequentially in order of the rows.

---